

On the Invariant Theory of Forms in n Variables

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§ 8. Reduction of the Folding Process to Fundamental Foldings.

Following Theorem VII, we now carry out the reduction of the folding process to fundamental foldings mentioned at the beginning of § 7. We call (cf. § 5) every folding of two cogredient rows S^ρ, T^ρ a “cogredient folding,” and specifically the cogredient folding of defect 1 of two rows S^ρ, T^ρ a “fundamental folding of type ρ .” We then prove, in completion of Theorem VI:

Theorem VIII. The general folding process (41) can be composed from the $(n - 1)$ fundamental foldings of type ρ , where $\rho = 1, 2, \dots, n - 1$. In other words, the successive application of the $n - 1$ fundamental foldings suffices to generate all invariant formations.

In the binary domain the folding process (41) coincides directly with the only possible fundamental folding; in the ternary domain I proved Theorem VIII in § 1 of my dissertation (cf. the Introduction). There, because of (44), the general cogredient foldings are still identical with the fundamental foldings, and it is noteworthy that Theorem VIII for n variables accomplishes the reduction to fundamental foldings, not merely the much weaker reduction to cogredient foldings. The proof is modeled in principle on the ternary proof: the most general foldings are constructed from the fundamental foldings by applying the symbolic identities. We shall generate:

1. arbitrary foldings of defect 1 by cogredient foldings of defect 1, that is, by fundamental foldings (the analogue of the ternary case),
2. arbitrary foldings of defect λ by cogredient foldings of defect λ and arbitrary foldings of defect 1,
3. cogredient foldings of defect λ by cogredient foldings of defect $\lambda - 1$ and arbitrary foldings of defect 1.

As is essential, we assume by Theorem VII that the two forms S and T to be folded are normal forms. Thus, by (46), the relations hold:

$$\begin{aligned} (q_{n-\sigma_1-\lambda} S^{\sigma_1} \mid S_{n-\sigma_2} p_{\sigma_2-\lambda}) &= 0, & (\sigma_1 \geq \sigma_2, \lambda = 1, 2, \dots, \sigma_2, n - \sigma_1), \\ (q_{n-\tau_1-\lambda} T^{\tau_1} \mid T_{n-\tau_2} p_{\tau_2-\lambda}) &= 0, & (\tau_1 \geq \tau_2, \lambda = 1, 2, \dots, \tau_2, n - \tau_1). \end{aligned} \quad (62)$$

It is further sufficient, by § 2 (end), to suppose that the forms S and T are linear in all variable rows; that is, the constructed forms belong to the form row

$$f = (S^1 \mid p_1)(S^2 \mid p_2) \cdots (S^{n-1} \mid p_{n-1})(T^1 \mid p_1)(T^2 \mid p_2) \cdots (T^{n-1} \mid p_{n-1}).$$

1) Generation of $(q_{n-\sigma-1} S^\sigma \mid T_{n-\tau} p_{\tau-1})$, where $\sigma > \tau$, from fundamental foldings of type $\tau + \alpha$, with $\alpha = 0, 1, \dots, \sigma - \tau$. From the fundamental folding of type τ

$$(q_{n-\tau-1} S^\tau \mid T_{n-\tau} p_{\tau-1})$$

we obtain, by the substitution $w \sim T_{n-\tau} p_{\tau-1}$, a form of the form row f with three rows of upper weight index (cf. § 1) $\tau + 1$:

$$(w S^\tau \mid p_{\tau+1})(S^{\tau+1} \mid p_{\tau+1})(T^{\tau+1} \mid p_{\tau+1}). \quad (63)$$

A fundamental folding of type $\tau + 1$ applied to (63) gives

$$(q_{n-\tau-2} w S^\tau \mid S_{n-\tau-1} p_\tau);$$

from (31), by the substitution $q_{n-\tau-2} w = q'_{n-\tau-1}$, this has the value

$$\varepsilon(q_{n-\tau-2} w S^{\tau+1} \mid S^\tau p_\tau) + \sum_{\alpha=1}^{\tau, n-\tau-1} \Pi(q'_{n-\tau-1-\alpha} S^{\tau+1} \mid S_{n-\tau} p_{\tau-\alpha}).$$

The expression under the summation sign vanishes identically by (62); thus we have reached a form

$$(w S^{\tau+1} \mid p_{\tau+2})(S^{\tau+2} \mid p_{\tau+2})(T^{\tau+2} \mid p_{\tau+2}),$$

which is obtained from (63) by changing τ into $\tau + 1$. Repeating the process $\sigma - \tau$ times therefore leads to the desired folding:

$$(wS^\sigma \mid p_{\sigma+1}) = \varepsilon(q_{n-\sigma-1}S^\sigma \mid T_{n-\tau}p_{\tau-1}).$$

We indicate the process just carried out briefly by the formula

$$T^\tau S^\tau, \quad S^{\tau+1}, \quad S^{\tau+2}, \dots, S^\sigma; \quad (64)$$

and see that only the normal-form property of S , not that of T , is used. A process dual to (64) can be performed using only the normal-form property of T , in the reverse order, namely:

$$S^\sigma T^\sigma, \quad T^{\sigma-1}, \quad T^{\sigma-2}, \dots, T^\tau, \quad (65)$$

according to the relations

$$\begin{aligned} (q_{n-\sigma-1}S^\sigma \mid T_{n-\sigma}p_{\sigma-1}) &= \varepsilon(T_{n-\sigma}yp_{\sigma-1}), & y &\sim q_{n-\sigma-1}S^\sigma, \\ (q_{n-\sigma}T^{\sigma-1} \mid T_{n-\sigma}yp_{\sigma-2}) &= \varepsilon(T_{n-\sigma+1}yp_{\sigma-2}), \\ &\vdots \\ (q_{n-\tau-1}T^\tau \mid T_{n-\tau-1}yp_{\tau-1}) &= \varepsilon(q_{n-\sigma-1}S^\sigma \mid T_{n-\tau}p_{\tau-1}). \end{aligned}$$

We should also point out the connection with the decrease of total weight given at the end of § 6: for foldings of defect 1 this decrease has the value $2(1 + (\sigma - \tau))$, and for fundamental foldings the value $2 \cdot 1$. Therefore, if composition from fundamental foldings is possible at all, exactly $\sigma - \tau + 1$ such foldings are necessarily required, as was carried out above.

2) Generation of general foldings from cogredient and fundamental foldings. The weight decrease for a general folding (41) is $2(\lambda^2 + \lambda(\sigma - \tau)) = 2[\lambda^2 + (\sigma - \tau)(1 + (\lambda - 1))]$. This gives the possibility of decomposing it into a cogredient folding of defect λ (weight decrease $2\lambda^2$) and $\sigma - \tau$ foldings of defect 1 with difference of weight indices $\lambda - 1$ (weight decrease $2\{1 + (\lambda - 1)\}$). In the case $\lambda = 1$ this decomposition agrees with the one given under 1); we show that it can indeed be realized.

From the cogredient folding of defect λ

$$(q_{n-\tau-\lambda}S^\tau \mid T_{n-\tau}p_{\tau-\lambda})$$

we obtain, by the substitution $R^\lambda \sim T_{n-\tau}p_{\tau-\lambda}$, a form with the two rows¹ of upper weight indices $\tau + \lambda$ and $\tau + 1$:

$$(R^\lambda S^\tau \mid p_{\tau+\lambda})(S^{\tau+1} \mid p_{\tau+1}). \quad (66)$$

The row with the lower weight index $\tau + 1$ belongs to a normal form; hence (66) admits a folding of defect 1 composed of λ fundamental foldings, in the order of (65):

$$(R^\lambda S^\tau)S^{\tau+\lambda}, \quad S^{\tau+\lambda-1}, \quad S^{\tau+\lambda-2}, \dots, S^{\tau+1}.$$

Taking account of (31) and (62), this gives

$$(q_{n-\tau-\lambda-1}R^\lambda S^\tau \mid S_{n-\tau-1}p_\tau) = \varepsilon(R^\lambda S^{\tau+1} \mid p_{\tau+\lambda+1}),$$

that is, a form obtained from (66) by replacing τ by $\tau + 1$, just as under 1) for (63). Repeating the process $\sigma - \tau$ times leads to the desired folding:

$$(R^\lambda S^\sigma \mid p_{\sigma+\lambda}) = \varepsilon(q_{n-\sigma-\lambda}S^\sigma \mid T_{n-\tau}p_{\tau-\lambda}).$$

In the whole process only the normal-form property of S was used. Viewing T as a normal form gives the dual process, with complete reversal of the order, according to the formulas

$$\begin{aligned} (q_{n-\sigma-\lambda}S^\sigma \mid T_{n-\sigma}p_{\sigma-\lambda}) &= \varepsilon(T_{n-\sigma}R_\lambda p_{\sigma-\lambda}), & R_\lambda &\sim q_{n-\sigma-\lambda}S^\sigma, \\ (q_{n-\sigma}T^{\sigma-1} \mid T_{n-\sigma}R_\lambda p_{\sigma-\lambda-1}) &= \varepsilon(T_{n-\sigma+1}R_\lambda p_{\sigma-\lambda-1}), \\ &\vdots \\ (q_{n-\tau-1}T^\tau \mid T_{n-\tau-1}R_\lambda p_{\tau-\lambda}) &= \varepsilon(q_{n-\sigma-\lambda}S^\sigma \mid T_{n-\tau}p_{\tau-\lambda}). \end{aligned}$$

¹**R823 source defect N04-S08-SRCDEF-004:** R823 line 4348 has the ungrammatical case ‘mit der beiden Reihen’, while the original print on p. 148 reads ‘mit den beiden Reihen’. The English follows the print as ‘with the two rows’.

3) Generation of cogredient foldings of defect λ from cogredient foldings of defect $\lambda - 1$ and fundamental foldings. The weight decrease for cogredient foldings of defect λ is $2\lambda^2 = 2((\lambda - 1)^2 + 2\lambda - 1)$, giving the possibility of a decomposition into a cogredient folding of defect $\lambda - 1$ and $2\lambda - 1$ fundamental foldings. This is most easily found by first putting $n = 2\lambda$. The only possible cogredient folding of defect λ is $(S^{n/2} \mid T_{n/2}) = (S^\lambda \mid T_\lambda)$; it contains one row u and one row x fewer than the folding of defect $\lambda - 1$ of the same rows, $(uS^\lambda \mid T_\lambda x)$. Conversely, we compose the folding of defect λ from fundamental foldings that effect the disappearance of a row u and a row x (weight decrease $2(n - 1) = 2(2\lambda - 1)$) and at the same time generate a new symbol row R^λ , together with a cogredient folding of defect $\lambda - 1$. The simplest folding that does this is the following one of defect 1:

$$(sT^\lambda \mid \sigma p_\lambda) = \varepsilon(S^{2\lambda-1} \mid R_{\lambda-1} p_\lambda) = \varepsilon(R^\lambda \mid p_\lambda),$$

where

$$R^{\lambda+1} = sT^\lambda, \quad s = S^1, \quad \sigma = S_1, \quad R^\lambda \sim \sigma R_{\lambda-1}.$$

By (65) and (64) it is composed of the $2\lambda - 1$ fundamental foldings

$$T^\lambda S^\lambda, S^{\lambda-1}, \dots, S^1; \quad R^{\lambda+1} S^{\lambda+1}, S^{\lambda+2}, \dots, S^{2\lambda-1}. \quad (67)$$

A folding of defect $\lambda - 1$, taking account of (21) and (62), gives

$$(uS^\lambda \mid \sigma R_{\lambda-1} x) = \varepsilon(R^{\lambda+1} \mid S_\lambda x) = \varepsilon(sT^\lambda \mid S^\lambda x) = \varepsilon(S^\lambda \mid T_\lambda).$$

The general case, for two rows S^ρ, T^ρ and arbitrary n , is obtained directly from the special one. In place of (67) one has the following $2\lambda - 1$ fundamental foldings:

$$T^\rho S^\rho, S^{\rho-1}, \dots, S^{\rho-\lambda+1}; \quad R^{\rho+1} S^{\rho+1}, S^{\rho+2}, \dots, S^{\rho+\lambda-1}. \quad (68)$$

The first half of the foldings gives the form

$$(q_{n-\rho-1} T^\rho \mid S_{n-\rho+\lambda-1} p_{\rho-\lambda}),$$

from which, by the substitutions

$$S_{n-\rho+\lambda-1} p_{\rho-\lambda} \sim w, \quad T^\rho w = R^{\rho+1},$$

and application of the second half of the foldings, one obtains

$$(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \mid R_{n-\rho-1} p_\rho).$$

Substituting

$$q_{n-\rho-\lambda} S^{\rho+\lambda-1} \sim y,$$

one obtains, by a cogredient folding of defect $\lambda - 1$,

$$(q_{n-\rho-\lambda+1} S^\rho \mid y R_{n-\rho-1} p_{\rho-\lambda+1}). \quad (69)$$

We show that this is the desired folding of defect λ . Substitute

$$R_{n-\rho-1} p_{\rho-\lambda+1} = R_{n-\lambda}$$

and observe that resolving y gives

$$(q_{n-\rho-\lambda+1} R^\lambda \mid S_{n-\rho} y) = \varepsilon(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \mid S_{n-\rho} p'_{\rho-1}) = 0.$$

Thus by (20) the value of (69) is

$$\varepsilon(S^\rho R^\lambda \mid p_{\rho+\lambda-1} y) = \varepsilon(q_{n-\rho-\lambda} S^{\rho+\lambda-1} \mid [S^\rho R^\lambda]_{n-\rho-\lambda} p_{\rho+\lambda-1}).$$

This folding is of type (43), hence equivalent to

$$(q_{n-\rho-\lambda} S^\rho \mid R_{n-\rho-1} p_{\rho-\lambda+1}) = \varepsilon(w T^\rho \mid R_\lambda p_{\rho-\lambda+1}), \quad R_\lambda \sim q_{n-\rho-\lambda} S^\rho.$$

Here R_λ is already of the desired final form;² the expression corresponds to the form $(sT^\lambda \mid S_\lambda x)$ in the special case. Now observe that resolving w and R_λ gives

$$(w R^{n-\lambda} \mid T_{n-\rho} p_{\rho-\lambda+1}) = \varepsilon(q'_{\lambda-1} R^{n-\lambda} \mid S_{n-\rho+\lambda-1} p_{\rho-\lambda}) = \varepsilon(q''_{n-\lambda-1} S^\rho \mid S_{n-\rho+\lambda-1} p_{\rho-\lambda}) = 0,$$

²R823 line 4442 splits the German compound *Schluß form*; the original print on p. 150 has *Schlußform* closed as one word. The printed lexical reading is followed; stable source-defect ID N04-S08-SRCDEF-001.

so that by (21) we obtain the value

$$(wq_{n-\rho+\lambda-1} \mid T_{n-\rho}R_\lambda) = \varepsilon(q_{n-\rho+\lambda-1}[T_{n-\rho}R_\lambda]^{n-\lambda} \mid S_{n-\rho+\lambda-1}p_{\rho-\lambda})$$

equivalent to $(q_{n-\rho-\lambda}S^\rho \mid T_{n-\rho}p_{\rho-\lambda})$.

In the whole process only the normal-form property of S , not that of T , was used. Thus the cogredient folding of defect $\lambda - 1$ used to generate (69) can be replaced by $2\lambda - 3$ fundamental foldings and a cogredient folding of defect $\lambda - 2$, which in turn admits a decomposition into fundamental foldings, and so on. Analogously, generation is obtained by using the normal-form property of T and applying the dual process, that is, the fundamental foldings³

$$S^\rho T^\rho, \quad T^{\rho+1}, \dots, T^{\rho+\lambda-1}; \quad R^{\rho-1} T^{\rho-1}, \quad T^{\rho-2}, \dots, T^{\rho-\lambda+1},$$

and a folding dual to (69). The folding

$$(q_{n-\rho-\lambda}T^\rho \mid S_{n-\rho}p_{\rho-\lambda}),$$

equivalent to the preceding one, is then obtained.

Adding the decomposition obtained under 2), we have indeed obtained a generation of the most general foldings from fundamental foldings. It remains to show that this generation is unique apart from permutation of the order; that is, that to each folding there corresponds one and only one number α_ρ of fundamental foldings of type ρ , where $\rho = 1, 2, \dots, n - 1$.

Let m_ρ be the degrees in the variable rows p_ρ of the initial form f , and l_ρ the degrees of the form produced by folding. Each fundamental folding of type ρ transforms the degrees $m_{\rho-1}, m_\rho, m_{\rho+1}$ respectively into $m_{\rho-1} + 1, m_\rho - 2, m_{\rho+1} + 1$. Thus the α satisfy the system of equations

$$m_\rho - l_\rho = -\alpha_{\rho-1} + 2\alpha_\rho - \alpha_{\rho+1}, \quad (\rho = 1, 2, \dots, n - 1), \quad (70)$$

where α_0 and α_n are to be set equal to zero, and whose determinant has the value n .⁴ The α are thereby uniquely determined and, by Theorem VIII, are positive integers; this gives conditions for the differences $m_\rho - l_\rho$.

The results of this paragraph hold by virtue of relations (62); however, if relations (62) do not hold, the final expressions⁵ are by no means equivalent to the initial expressions. The definition of equivalence given in § 6 also admits the formulation:⁶ “Two forms are equivalent if and only if they differ by forms of higher folding, that is, by forms that exhibit a greater weight decrease.” This greater weight decrease always occurs when the two rows $q_{n-\sigma-\lambda}, p_{\tau-\lambda}$ entering the folding are really variable rows, as happens in (41), (42), in the foldings appearing in this paragraph as equivalent, and also in the equivalent forms of Theorem VII. But it need not occur when these rows are derived from symbol rows by substitution, as was the case for the forms of this paragraph that could be expressed in terms of one another by virtue of (62). One easily sees that some of the vanishing forms even have a higher total weight.

³**R823 source defect N04-S08-SRCDEF-003:** R823 line 4457 gives the first dual sequence with descending exponents $T^{\rho-1}, \dots, T^{\rho-\lambda+1}$, whereas the original print on p. 150 has the ascending sequence $T^{\rho+1}, \dots, T^{\rho+\lambda-1}$. The English restores the printed sequence. The second sequence in the display remains descending, as printed.

⁴If Δ_n denotes the n -rowed determinant, then the recurrence formula holds: $\Delta_n = 2\Delta_{n-1} - \Delta_{n-2}$; that is, $\Delta_n - \Delta_{n-1} = \Delta_{n-1} - \Delta_{n-2} = \dots = \Delta_2 - \Delta_1 = 1$, since $\Delta_2 = 3$ and $\Delta_1 = 2$.

⁵R823 line 4475 splits the German compound *Schluß ausdrücke*; the original print on p. 151 has *Schlußausdrücke* closed as one word. The printed lexical reading is followed; stable source-defect ID N04-S08-SRCDEF-002.

⁶Cf. the consideration at the end of the next paragraph.